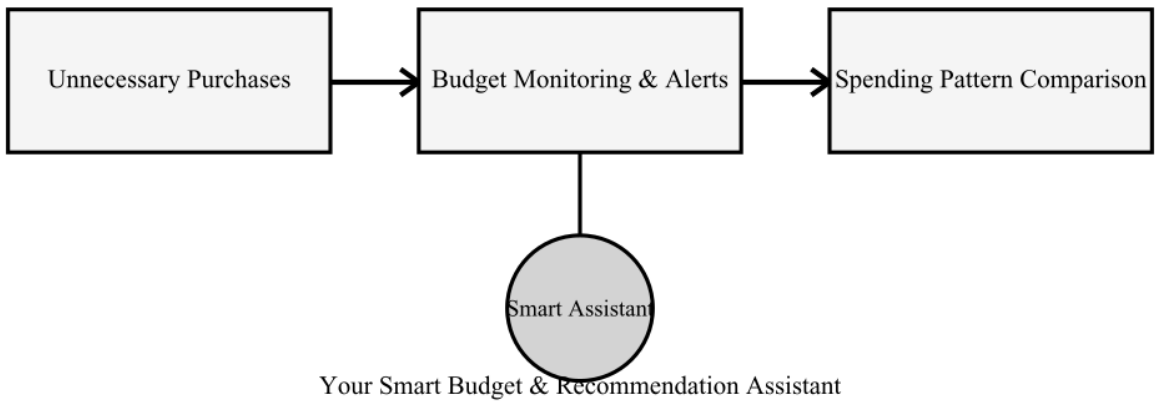
YOUR SMART BUDGET & RECOMMENDATION ASSISTANT

1.2 Unplanned Spending

Expanded Problem Statement



Your Smart Budget & Recommendation Assistant

Overview

Unplanned purchases can gradually increase monthly expenses. A user may make frequent purchases for shopping, entertainment, food delivery, online services, or other non-essential needs without checking the remaining budget. The problem becomes more serious when many small purchases combine into a large amount by the end of the month.

Unnecessary Purchases

The assistant can highlight spending that is outside the user's planned budget or that has increased significantly compared with previous periods. This does not prevent the user from spending; instead, it provides information that can help the user make a conscious decision.

Budget Monitoring & Alerts

Each category can have a predefined spending limit. When spending approaches the limit, the system can provide an alert. If the limit is exceeded, a stronger notification can explain which category caused the budget to go over.

Spending Pattern Comparison

The system can compare current spending with historical spending. For example, if shopping expenses are much higher than the previous month, the assistant can identify the increase and recommend reviewing that category.

1.2 Unplanned Spending – Implementation

Sample Coding Logic

The following simple Python example demonstrates the core logic related to this problem area. It can later be integrated with a database, dashboard, notification system, and predictive model.

```
budget_limit = 5000
current_spending = 5600

if current_spending > budget_limit:
excess = current_spending - budget_limit
print("Budget exceeded by:", excess)
print("Alert: Review non-essential spending")
else:
print("Spending is within budget")
```

System Benefits

The feature is designed to make financial management easier, more organized, and more understandable. By converting raw expense information into useful insights, the assistant can help users take timely action.

1.2 Unplanned Spending – Summary

Key Points

Area	Current Challenge	Proposed Support
Unplanned purchases	Small purchases accumulate	Monitor category spending.
Budget limits	Users may exceed planned amounts	Set limits and trigger alerts.
Unexpected increases	Spending may rise without notice	Compare current and previous patterns.

Expected Outcome

The assistant can improve spending awareness by showing users when their actual expenses differ from their plan. Timely alerts and category-level information can help users review non-essential spending before it significantly affects the monthly budget.